ENGINEERS WALLAH

CHEMISTRY

'Brief'

Lecture XII

Organic Chemistry



By: Sunil Bhaiya

TOPICS *to be covered*

1. Important Facts About Carbon
2. Introduction to Covalent Bond
3. Versatile Nature of Carbon
4. Allotropes of Carbon
5. Introduction to Hydrocarbons
6. Functional Groups and IUPAC Nomenclature
7. Important Carbon Compounds

Knowledge Ride On



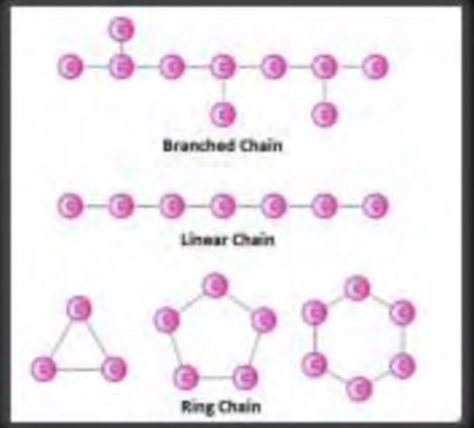
Important Facts About Carbon

Knowledge Ride On



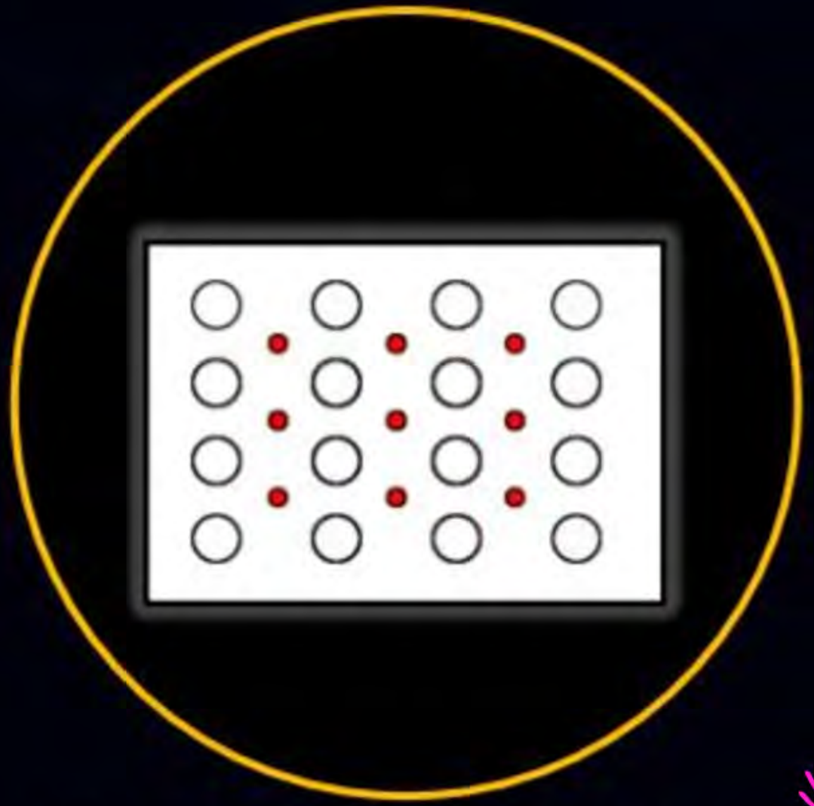
Introduction to Covalent Bond

Knowledge Ride On



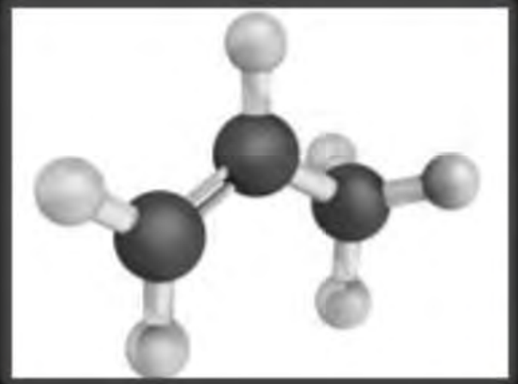
Versatile Nature of Carbon

Knowledge Ride On



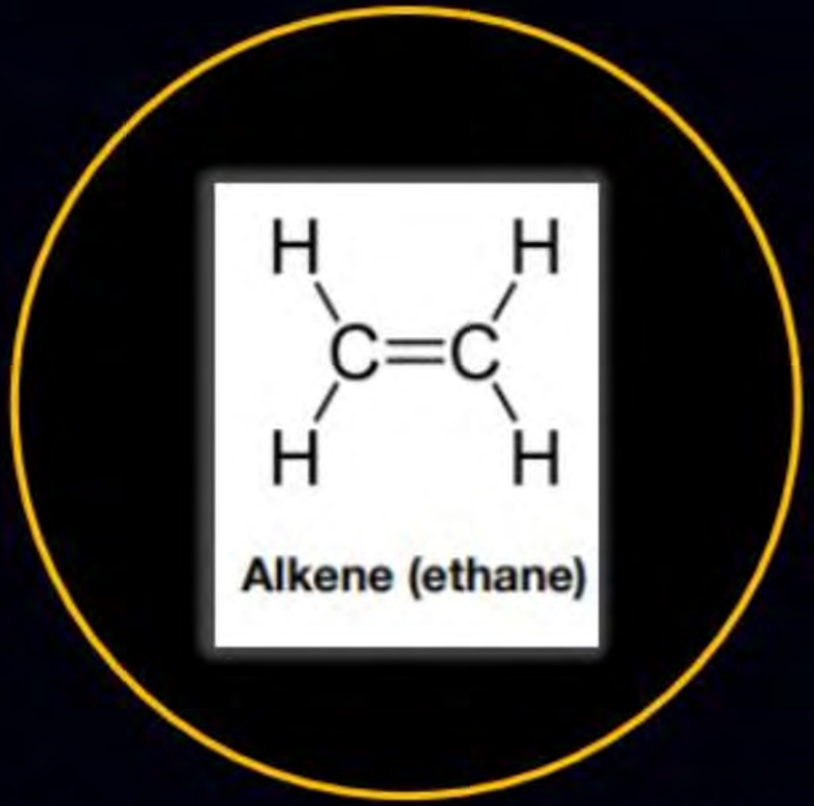
Allotropes of Carbon

Knowledge Ride On



Introduction to Hydrocarbons

Knowledge Ride On



'BRIEF'

Functional Groups and IUPAC Nomenclature

Knowledge Ride On



Important Carbon Compounds

Knowledge Ride On



Efficiency Hacks

Glossary

Covalent Bond (सहसंयोजक बंधन)
Allotropes (अपररूप)
Catenation (श्रृंखलन)
Functional group (कार्यात्मक समूह)
Soap (साबुन)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

(v)

Siderite is an ore of:

SSC JE 2019

Another name → chalybite → $FeCO_3$

A Copper

B Aluminium

C Zinc

D Iron ✓

Another Ores of iron:

(i) Haematite → Fe_2O_3

(ii) Magnetite → Fe_3O_4 [$FeO.Fe_2O_3$]

(iii) Iron pyrite → FeS_2

(iv) limonite → $2Fe_2O_3.3H_2O$

Question

Dolomite is an ore of:

$MgCO_3 . CaCO_3$ OR $[CaMg(CO_3)_2]$

SSC JE 2012

A Aluminium

B Magnesium

C Copper

D Lead

Question

$H_2O + O_2 + CO_2$

A vessel is exposed to moist air for a long time. It acquires a dull, green coating. What is the vessel made up of?

SSC JE 2022

Patina → $CuCO_3 \cdot Cu(OH)_2$ → Basic copper carbonate

A Iron

B Gold

C Copper

D Phosphorus

Concept Bites – Important Facts About Carbon

Important Facts About Carbon

(iv) Human body → 18.1% Carbon

Lithosphere → Earth's crust (Abundance)

I st → Oxygen (Non-metal)

III rd → Aluminium (Metal)

XV th → Carbon (Non-metal)

(i) Carbon is the **15th most abundant** element in the earth's crust. [LITHOSPHERE]

(ii) Earth's crust has **.02% carbon** in form of minerals like carbonates, hydrogen carbonates, coal and petroleum.

Atmosphere

(i) Nitrogen [78.08%]
(ii) Oxygen [20.95%]
(iii) Argon [.93%]
IV Carbon dioxide [.03%]
V Rest gases [.01%]

(iii) Carbon is present in atmosphere primarily in the form of carbon dioxide. Its concentration is **0.03%**.

↳ Traces of carbon monoxide (CO)

Concept Bites – Introduction to Covalent Bond

Covalent Bond (सहसंयोजक बंधन)

→ non-metals
→ sharing of electron(s)

Element	Atomic or proton number	Number of electrons
Hydrogen (H)	1	1

K
1

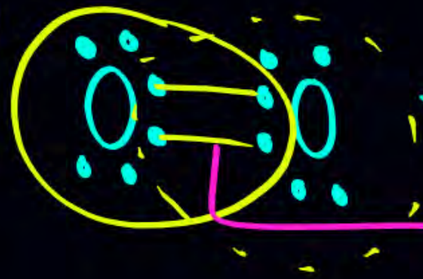


→ H_2

→ single covalent bond

Element	Atomic or proton number	Number of electrons
Oxygen (O)	8	8

K L
2 6



→ O_2

→ double covalent bond

Element	Atomic or proton number	Number of electrons
Nitrogen (N)	7	7

K L
2 5



→ N_2

→ triple covalent bond

MONOVALENT
each H atom shares 1 e⁻

DIVALENT

TRIVALENT

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following correctly defines a covalently bonded nitrogen molecule?

A N_2, Trivalent

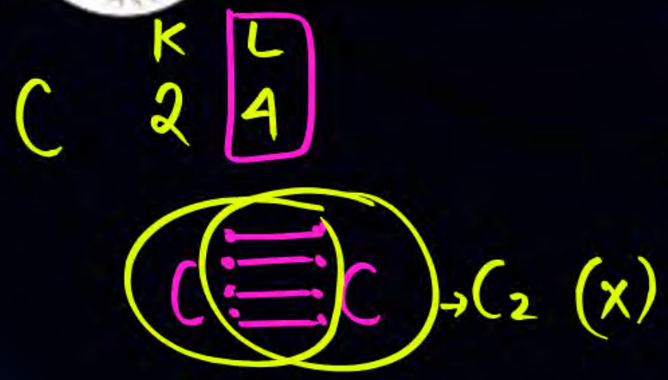
B N_3, Monovalent

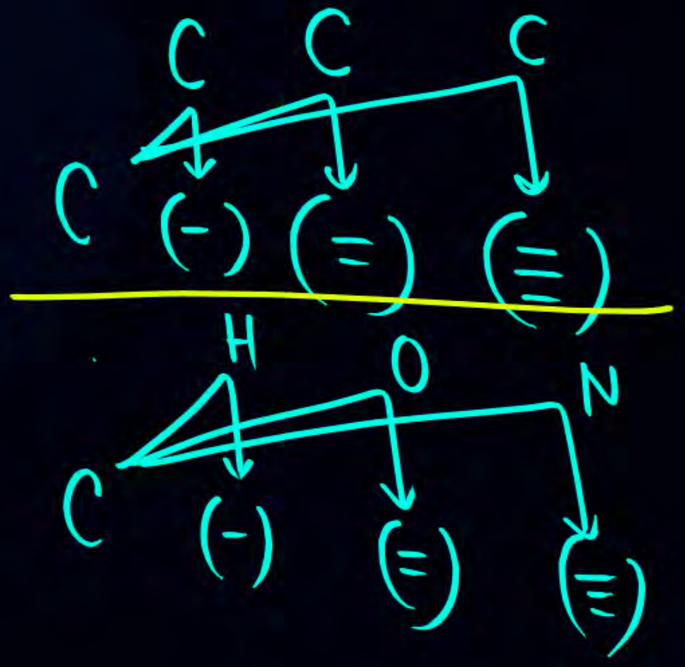
C N_2, Divalent

D N_3, Trivalent

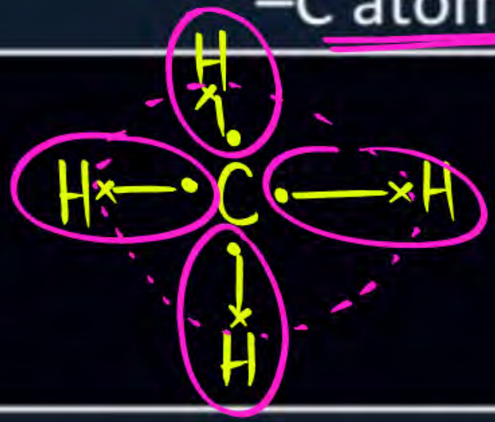
Concept Bites – Versatile Nature of Carbon

Versatile Nature of Carbon





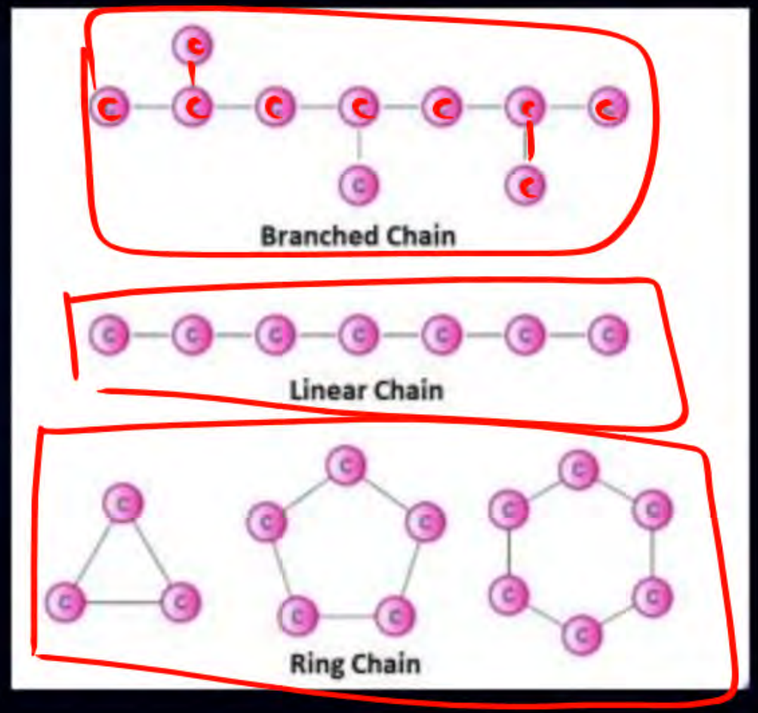
(i) Each –C atom shares 4 electrons to attain stable noble gas configuration. It shows ***tetravalency*** and –C atom is ***tetravalent***.



(ii) Carbon has a tendency to form ***multiple bonds*** like single, double and triple with different elements.

Versatile Nature of Carbon

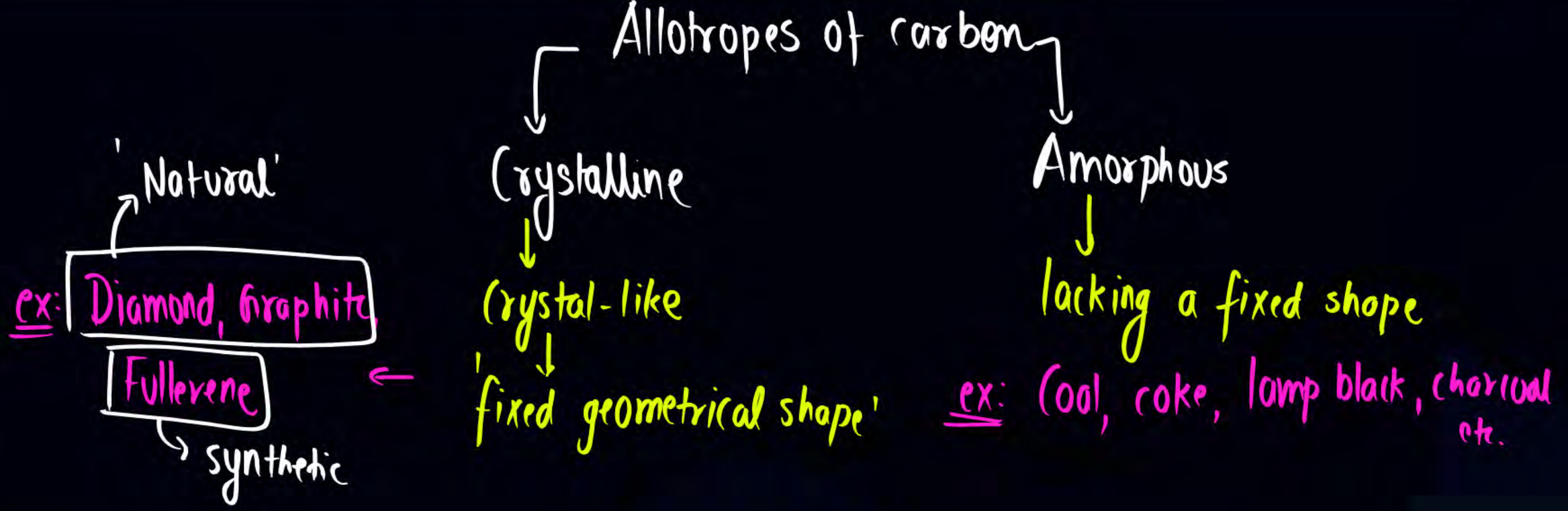
(iii) Carbon atom links with another carbon atoms to form long, branched and closed carbon rings. *This self-linking property is known as* ***catenation*** ***(श्रृंखलन)***.



Concept Bites – Allotropes of Carbon

Allotrope (अपररूप)

Same element exists in the different physical structures while being in the same physical state are called allotropes and this phenomena is known as ***allotropy*** **or** ***allotropism***.



Diamond

(i) Diamond is the ***hardest naturally occurring substance*** and the ***best conductor of heat***.

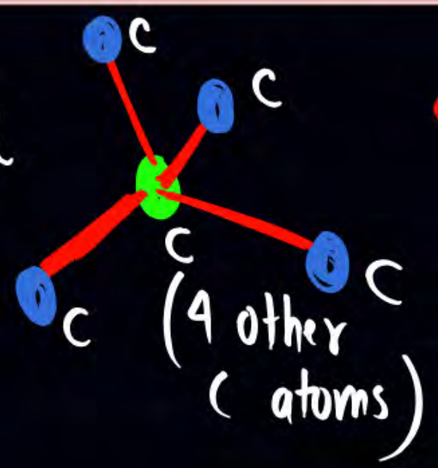
(ii) In diamond, each –C atom is bonded to 4 other carbon atoms leaving no free electron. Tetrahedral units combine to form a 3D rigid structure of diamond.

(ii)' 8 atoms [one at each corner]

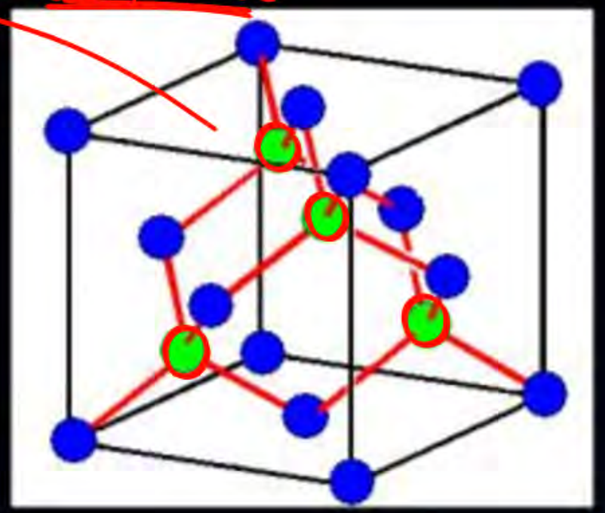
6 atoms [one at each face]

4 atoms [In btw]

→ Tetrahedral Unit



UNIT CELL



Graphite

'Graphite'

(i) Graphite is an ***opaque***, ***greyish black***, ***brittle*** and ***lustrous form of C***.

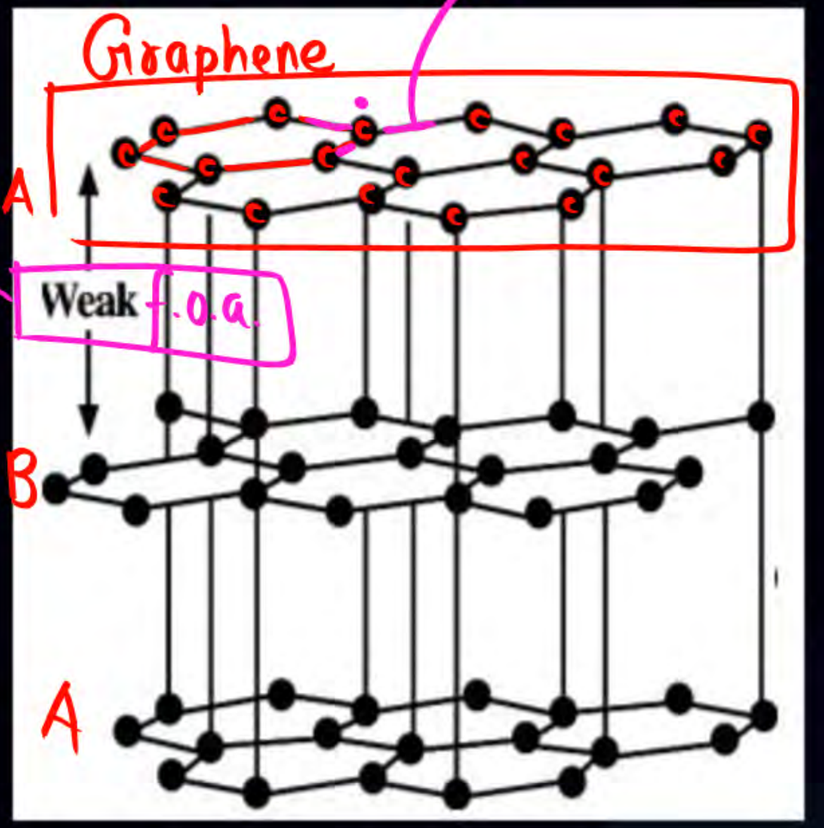
(ii) Each layer of graphite is called ***Graphene*** and alternate layers are same. Atoms are arranged in layers of ***hexagonal array***.

(iii) In graphite, each –C atom is bonded to 3 other carbon atoms leaving a free electron. Hence, graphite conducts electricity.

'layers slide over each other'

→ Soft, smooth & slippery

→ Solid lubricant



Vanderwaal f.o.a.

Fullerene

(i) The first fullerene (C_{60}) was discovered in 1985 by H.W. Kroto, R. E. Smalley and R. F. Curl and was named after a US architect Buckminster Fuller as the structure of C_{60} was similar to the geodesic dome designed by him.

(ii) It contains 20 six membered rings and 12 five membered rings. The six membered rings make bonds with five and six membered rings while five membered rings only make bonds with six membered rings. *The structure is also known as* Soccer ball-like or bucky ball structure.



Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following is the structure of graphite?

SSC JE 2017

A. Rigid three-dimensional [Diamond]

B. Hexagonal array [Graphite]

C. Like football [Fullerene]

D. No option is correct

Question

In diamond, every carbon atom is bonded to __4__ other carbon atoms.

A 3 (In case of graphite)

B 2

C 4

D 5

Question

The pencil lead is made from which of the following element?

A Silicon

B Carbon (Graphite)

C Lead

D Sodium

Concept Bites – Introduction to Hydrocarbons

Hydrogen

Carbon

Introduction to Hydrocarbons

	Alkane	Alkene	Alkyne
General formula *(Defined for one double and triple bond in a hydrocarbon)*	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
Covalent bond between –C atoms	Single Covalent Bond [Btw. C atoms]	Double Covalent Bond [One btw. two C atoms]	Triple Covalent Bond [One btw. two C atoms]
Type of hydrocarbon	Saturated H.C.	← Unsaturated H.C. →	
Suffix added → phrase after	- ane	- ene	- yne

ALKANE $[C_nH_{2n+2}]$ $n=2 \rightarrow C_2H_6$

ALKENE $[C_nH_{2n}]$ $n=3 \rightarrow C_3H_6$

ALKYNE $[C_nH_{2n-2}]$ $n=4 \rightarrow C_4H_6$

ROOT WORD + SUFFIX

Eth + ane → Ethane

Prop + ene → Propene

But + yne → Butyne

C-I

Root Word

Chain length (or) no. of carbon atoms		Root word
C_1	→	Meth-
C_2	→	Eth-
C_3	→	Prop-
C_4	→	But-
C_5	→	Pent-
C_6	→	Hex-
C_7	→	Hept-
C_8	→	Oct-
C_9	→	Non-
C_{10}	→	Dec-

C-II

ALKANE $[C_nH_{2n+2}]$

$n=1$

CH_4

Methane

ALKENE $[C_nH_{2n}]$

CH_2 ✗

ALKYNE $[C_nH_{2n-2}]$

$CH_0 \rightarrow C$ ✗

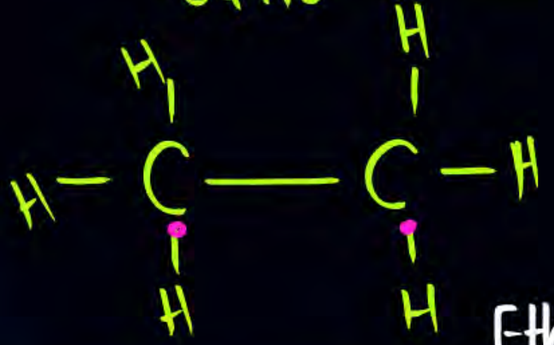
The first member of alkene and alkyne family is ethene and ethyne because minimum 2 C atoms are required for a double/triple covalent bond.

ALKANE

C_nH_{2n+2}

$n=2$

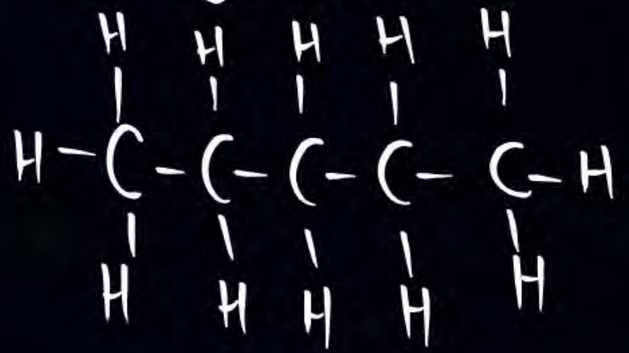
C_2H_6



Ethane

$n=5$

C_5H_{12}



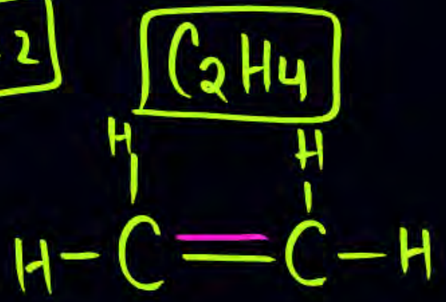
Pentane

ALKENE

C_nH_{2n}

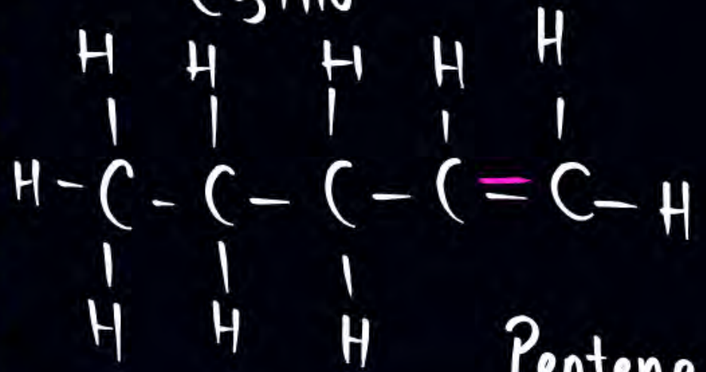
$n=2$

C_2H_4



Ethene

C_5H_{10}

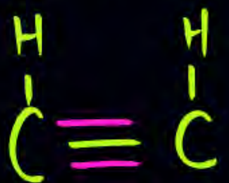


Pentene

ALKYNE

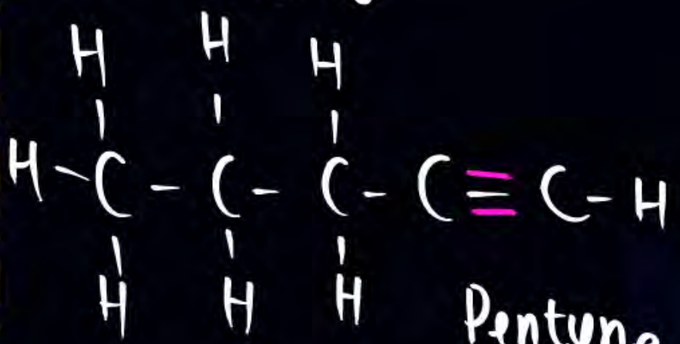
C_nH_{2n-2}

$n=2$, C_2H_2



Ethyne

C_5H_8



Pentyne

Trick to Calculate Number of Covalent Bonds

Alkanes	No. of C + no. of H - 1	(Butane) C_4H_{10}	$4 + 10 - 1 = 13$
Alkenes	No. of C + no. of H	(Butene) C_4H_8	$4 + 8 = 12$
Alkynes	No. of C + no. of H + 1	(Butyne) C_4H_6	$4 + 6 + 1 = 11$

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which is a colourless, odourless gas of the alkane series of hydrocarbons with chemical formula C_3H_8?

SSC JE 2022

Prop + ane
[Root word] [Suffix]

A Propane (C_3H_8)

B Pentane (C_5H_{12})

C Butane (C_4H_{10})

D Ethane (C_2H_6)

Concept Bites – Functional Groups and IUPAC Nomenclature

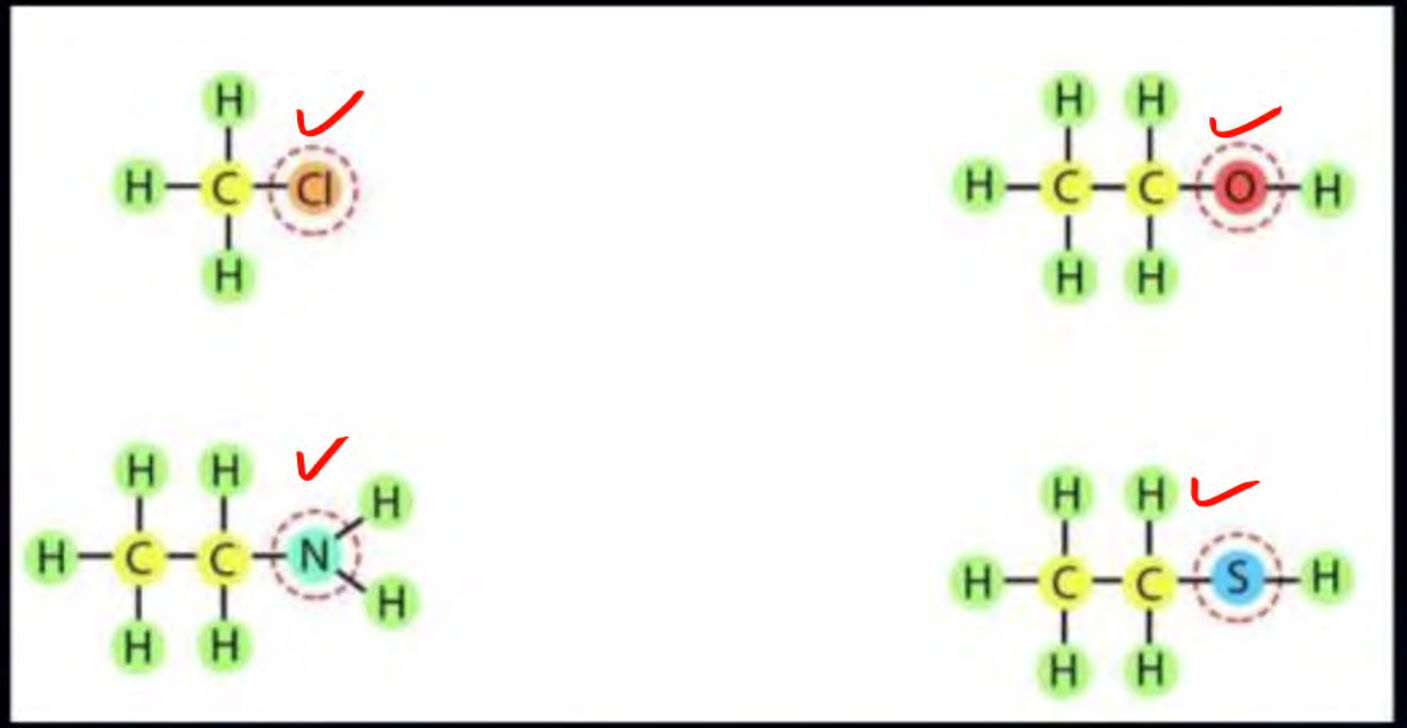
'BRIEF'

Heteroatom/Functional Group (कार्यात्मक समूह)

↳ 'different atom'

↳ 'provides function to the compound'

Hydrogen atom/atoms can be replaced by any other elements like Cl, S, N, O and provides completely different properties.





Give a Thought

What is an alkyl group?!

Represented by 'R'

$-C_nH_{2n+1}$

$(n=1)$	$-CH_3$	methyl
$(n=2)$	$-C_2H_5$	ethyl
$(n=3)$	$-C_3H_7$	propyl

Heteroatom/Functional Group (कार्यात्मक समूह)

Phrase after

Phrase before

Hetero atom	Class of compounds	Formula of Functional Group	Prefix/Suffix
F, Cl, Br, I	Halo- (Chloro/bromo) alkane / alkyl halides	X: F, Cl, Br, I (F.G.) R-X / $C_nH_{2n+1}-X$	Prefix → Fluoro, Chloro, Bromo, Iodo; no suffix
Oxygen	1. Alcohol	Hydroxyl F.G. R-OH / $C_nH_{2n+1}-OH$	Suffix → ol
	2. Aldehyde	aldehydic F.G. R-CHO / $C_nH_{2n+1}-CHO$	Suffix → al
	3. Ketone	Carbonyl F.G. $R-\overset{O}{\overset{\|\|}{C}}-R'$ / $C_nH_{2n+1}-\overset{O}{\overset{\|\|}{C}}-C_mH_{2m+1}$	Suffix → one
	4. Carboxylic acid	Carboxyl F.G. R-COOH / $C_nH_{2n+1}-COOH$	Suffix → oic acid

Heteroatom/Functional Group (कार्यात्मक समूह)

Heteroatom	Formula of Functional Group	Prefix/Suffix
Oxygen (ESTER)	 Carboalkoxy F.G.	Suffix: oate
Oxygen (ETHER)	 Ether F.G.	Prefix: Alkoxy

IUPAC Nomenclature of Organic Compounds

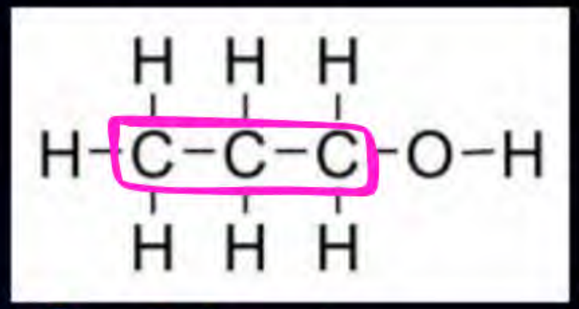
Prefix + Root Word + Primary suffix + Secondary suffix

Prefix → Side chain / branch / substituent / halogen / ether

Root Word → no. of C atoms (main C chain)

Primary suffix → -one, -ene, -yne

Secondary suffix → functional group



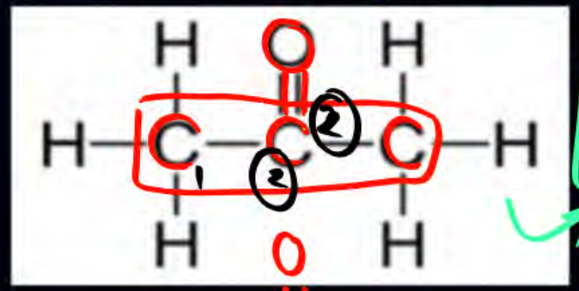
(I) Select longest C chain containing f.g. if it contains C.

NO

(II) Prop + ane + ol

Propanol OR Propan-1-ol OR 1-propanol

(III) After 'e' if 'a', 'i', 'o', 'u', 'y' comes; eliminate 'e'.



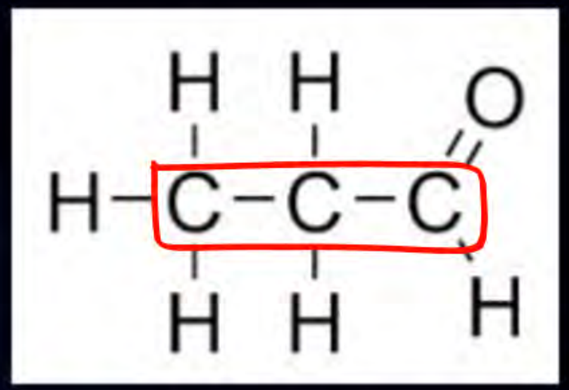
C_3H_6O

Common name → 'Acetone'

Prop + ane + one

Propan-2-one OR 2-propanone

IUPAC Nomenclature of Organic Compounds



Prefix + R.W + P.S. + S.S.

Prefix → –

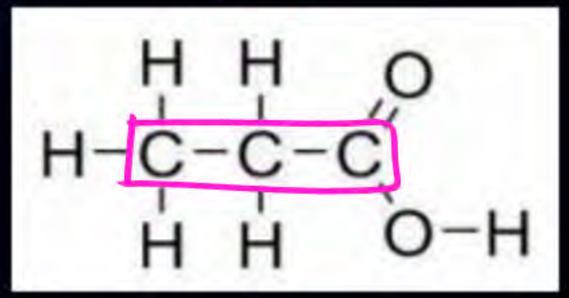
R.W → Prop

P.S. → ane + al

Prop + ane + al

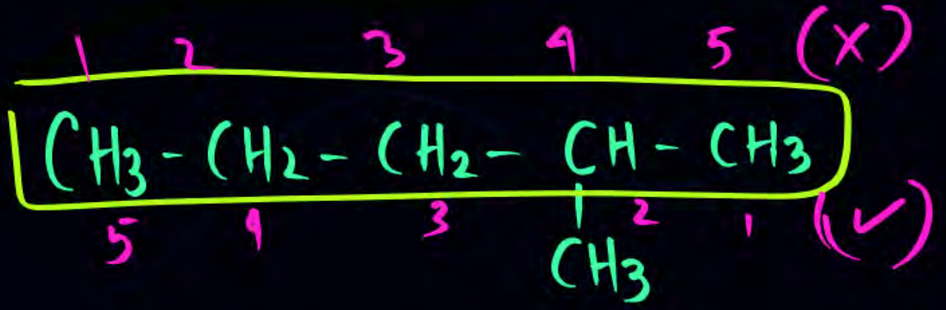
Propanal

R-COOH



– + Prop + ane + oic acid

Propanoic acid

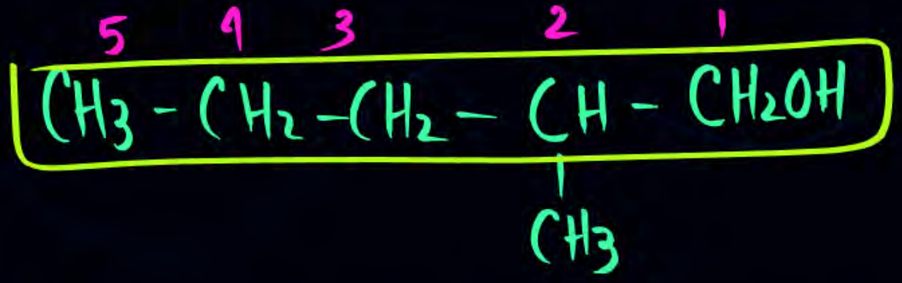


(I) Select longest C chain

(II) Start giving side chain / substituent / branch lowest no. possible

(III) Prefix + R.W. + P.S. + S.S.

= 2-methylpentane



(I) Select longest C chain containing F.G

(II) Give F.G. lowest no. as possible.

(III) Prefix + R.W. + P.S. + S.S.

2-methyl pent ~~ane~~ ol

2-methylpentan-1-ol

How would you name the following compounds?

$CH_3—CH_2—Br$

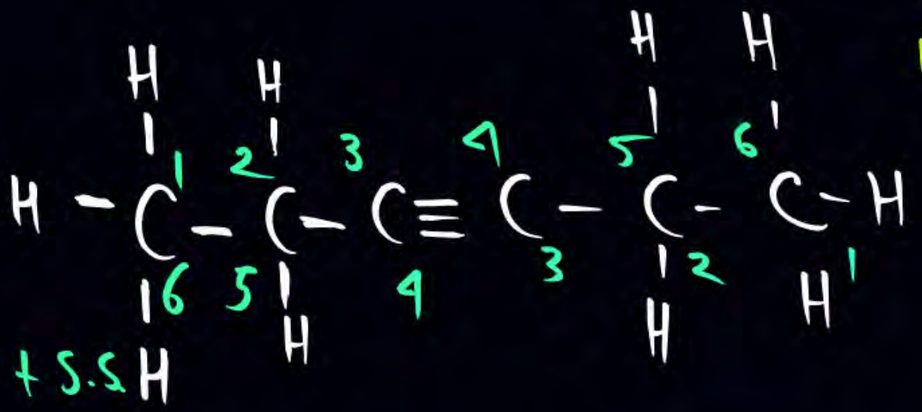
$C_2H_5 - Br$

(R-X)

Prefix + R.W. + P.S.

↓

Bromoethane



Prefix + R.W. + P.S. + S.S.

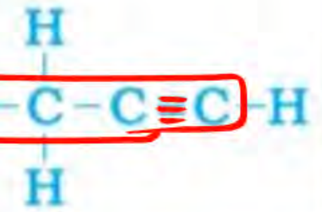
—

↓ ↓

hex yne

—

↳ hex-3-yne



C_6H_{10}

Prefix + R.W. + P.S. + S.S.

↓ ↓ ↓ ↓

NO hex yne NO

↓

hexyne or hex-1-yne



R-O-R'

→ Consider smaller alkyl chain as substituent

→ larger alkyl group → consider as parent chain

→ Alkoxy alkane
↓ ↓
S.C. M.C.

Prefix + R.W + P.S. + S.S.
↓ ↓ ↓ ↓
— hept + ane ol

heptan-2-ol

Prefix + R.W + P.S. + S.S.
↓ ↓ ↓
2-methoxy + hept + ane —

2-methoxyheptane

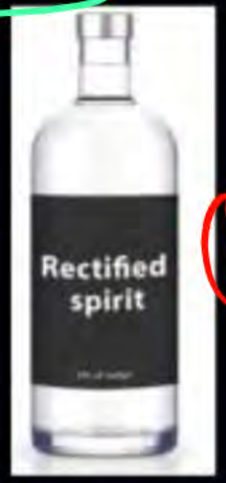
Concept Bites – Important Carbon Compounds

$R\text{-}OH$ / $C_nH_{2n+1}\text{-}OH$ → C_2H_5OH / CH_3CH_2OH

Some Important Terminologies of Ethanol → 'OH'

- 2 carbon
- single covalent bond

Absolute ethanol



(I) 95% ethanol + 5% water

Used as 'antiseptic' ← 'kills germs'



(II) 100% ethanol



(III) Ethanol + Blue dye + Methanol/Pyridine/Copper sulphate

To give indication that it is unfit for drinking

Poisonous substances ←

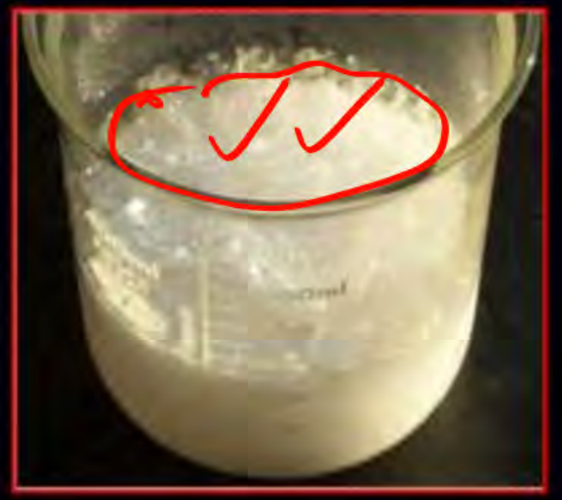
$C_nH_{2n+1}COOH$

An Important Terminology of Ethanoic acid/Acetic acid → CH_3COOH

(I) Vinegar (सिरका) → 5-8% solution of acetic acid in water.

(II) When acetic acid is cooled below 17 degree Celsius, it forms glacier-like crystals and hence known as glacial acetic acid.

Freezing point



Cleansing Agents

DETERGENTS

SOAPS

Sodium or potassium salts of long chain carboxylic acids

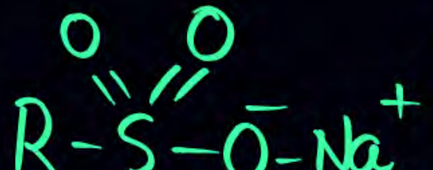
- $C_{17}H_{35}COOH$ (Stearic acid)
- $C_{15}H_{31}COOH$ (Palmitic acid)
- $C_{17}H_{33}COOH$ (Oleic acid)

GENERAL FORMULA → $R\text{-}COO^-Na^+$ / $RCOO^-K^+$

Sodium / potassium salt of long chain carboxylic acid

Soapless Soap

Sodium salt of long chain sulphonic acids



OR

RSO_3Na

Cleansing action is similar to soap.

ex: $C_{17}H_{35}COONa$ (Sodium stearate)

Imp. to Remember

→ First organic compound made in laboratory → Made by Wohler → 'Urea'

$$\underset{C\ A}{NH_4Cl} + \underset{C\ A}{KCNO} \xrightarrow{heat} NH_4CNO + KCl$$

← Inorganic compounds →

NH_4CNO ↓ rearrangement on heating

NH_2CONH_2 'Urea'

→ Benzene $[C_6H_6]$
↓
(August Kekule)

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following options correctly defines the concentration of vinegar and its IUPAC name?

A. 5–8% acetic acid in alcohol, Ethanoic acid

B. 5-8% acetic acid in water, Methanoic acid

C. 5–8% acetic acid in water, Ethanoic acid

D. 5–8% acetic acid in alcohol, Methanoic acid

Question

Which base is generally found in soaps?

→ Sodium salt / potassium salt

SSC JE 2017

A Calcium hydroxide

B Ammonium hydroxide

C Sodium hydroxide

D Magnesium hydroxide

Efficiency Hacks!

Tips to Follow During Any Competitive Exam

- Don’t stuck on one question. Save time! ✓
- Drink water during and before the exam. ✓
- Had a heavy breakfast/lunch 2 hours before the exam. ✓




#sbsathhai
#pwsathai
Thank
You !